NW Mock 2025 Chemistry Paper 2: Comprehensive Solution Guide

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Introduction

Welcome, dear students and future chemists! This guide is meticulously prepared to help you navigate the complexities of the **North West Regional Mock 2025 Chemistry Paper 2**. Chemistry is not just about memorizing formulas; it is about understanding the fundamental behavior of matter. In this document, I provide not just the answers, but the “why” behind them.

How to use this guide:

1. **Attempt first:** Try solving each question before looking at the solution.
2. **Understand the Concept:** Read the detailed explanations to grasp the underlying principles.
3. **Analyze the Wrong Options:** Pay attention to why other paths are incorrect to avoid common pitfalls in the GCE.
4. **Practice Marketing:** If you find these explanations helpful, remember that I am available for both **online and offline classes in Bamenda**. You can also find video explanations on my YouTube channel.

SECTION A: PHYSICAL AND GENERAL CHEMISTRY

Question 1(a)

Rewrite: Define the following terms as applied to the atom: (i) A molecule, (ii) Mass number. **Topic:** Atomic Structure and Bonding. **Detailed Concept:**

- **Molecule:** A molecule is the smallest particle of a substance (element or compound) that can exist independently and still retain the chemical properties of that substance. It consists of two or more atoms chemically bonded together.
- **Mass Number:** This is the total number of protons and neutrons (nucleons) present in the nucleus of an atom of an element. It is denoted by the letter 'A'. **Answer:** (i) A molecule is a group of two or more atoms held together by chemical bonds. (ii) Mass number is the sum of protons and neutrons in an atom's nucleus. **Why others are wrong:**
- Defining a molecule simply as "a group of atoms" is insufficient; they must be chemically bonded.
- Confusing Mass Number with Atomic Number (which is only the number of protons) is a frequent error.

Question 1(b)

Rewrite: How many protons and neutrons are contained in $^{59}_{27}\mathrm{Co}$? **Topic:** Subatomic Particles. **Detailed Concept:** In the notation $^{A}_{Z}X$, Z is the atomic number (protons) and A is the mass number (protons + neutrons). Therefore, Neutrons = $A - Z$. **Answer:**

- Protons = 27
- Neutrons = 59 - 27 = 32 **Why others are wrong:** Some students might subtract the atomic number from the mass number incorrectly or swap the values, leading to 59 protons, which is impossible for Cobalt.

Question 1©

Rewrite: Considering an atom, when is an electron said to be: (i) In the ground state, (ii) In the excited state? **Topic:** Quantum Mechanics and Electron Configuration. **Detailed Concept:** Electrons occupy specific energy levels. The most stable arrangement is the one with the lowest energy. **Answer:** (i) **Ground state:** When the electron is in the lowest possible energy level available to it. (ii) **Excited state:** When the electron has absorbed energy and jumped to a higher energy level. **Why others are wrong:** "Ground state" does not mean the electron is on the ground; it refers to energy levels. "Excited" does not mean the electron is moving faster, but that it is at a higher potential energy.

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Question 1(d)

Rewrite: Complete the table by giving the shapes and bond angles of $\mathrm{SO_4^{2-}}$ and $\mathrm{NH_2^-}$. **Topic:** VSEPR Theory (Valence Shell Electron Pair Repulsion). **Detailed Concept:**

- $\mathrm{SO_4^{2-}}$ has 4 bonding pairs and 0 lone pairs on Sulfur, leading to a tetrahedral electron geometry and molecular shape.
- $\mathrm{NH_2^-}$ has 2 bonding pairs and 2 lone pairs on Nitrogen. The two lone pairs exert greater repulsion than bonding pairs, compressing the bond angle. **Answer:** | Species | Shape | Bond Angle | | :— | :— | :— | | $\mathrm{SO_4^{2-}}$ | Tetrahedral | 109.5° | | $\mathrm{NH_2^-}$ | Bent / V-shaped | ~104.5° | **Why others are wrong:**
- $\mathrm{SO_4^{2-}}$ is sometimes incorrectly drawn as square planar or trigonal planar if the octet rule is strictly followed without considering expanded octets for sulfur.
- $\mathrm{NH_2^-}$ is often mistaken for linear if lone pair repulsion is ignored. The two lone pairs on Nitrogen push the bonds closer together, reducing the angle from the ideal tetrahedral angle of 109.5° to approximately 104.5°.

Question 1(e)

Rewrite: (i) Explain the terms acid and conjugate base as used in the Bronsted-Lowry theory. (ii) A 0.010M solution of $\mathrm{NH_3}$ has a pH of 10.6 at 25°C. Calculate the concentration of $\mathrm{OH^-}$ ions. **Topic:** Acids and Bases. **Detailed Concept:**

- **Bronsted-Lowry Theory:** Defines acids and bases based on proton transfer. An acid donates a proton (H^+), and a base accepts a proton. When an acid donates a proton, the species remaining is its conjugate base.
- **pH and pOH relationship:** At 25°C, $pH + pOH = 14$. The concentration of hydroxide ions can be calculated from pOH using the formula $[\mathrm{OH^-}] = 10^{-pOH}$. **Answer:** (i) **Acid:** A substance that donates a proton (H^+). **Conjugate Base:** The species formed after an acid has donated a proton. (ii) Given $pH = 10.6$. $pOH = 14 - pH = 14 - 10.6 = 3.4$. $[\mathrm{OH^-}] = 10^{-pOH} = 10^{-3.4} = 3.98 \times 10^{-4}\ \mathrm{mol/dm^3}$. **Why others are wrong:**

- Using the Arrhenius definition (releasing H^+ in water) instead of the Bronsted-Lowry (proton donor) will lose marks here, as the question specifically asks for Bronsted-Lowry.
- A common mistake in calculation is to directly use 10^{-pH} to find $[\mathrm{OH}^-]$ or to forget the $pH + pOH = 14$ relationship, leading to incorrect values.

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Question 2(a)

Rewrite: (i) What is a standard solution? (ii) Determine the concentration of a $\mathrm{KMnO_4}$ solution used to titrate $25\ \mathrm{cm^3}$ of $2.388\mathrm{M}$ acidified sodium oxalate ($\mathrm{Na_2C_2O_4}$), using $18\ \mathrm{cm^3}$ of $\mathrm{KMnO_4}$. The equation for the reaction is given below: $2\mathrm{MnO_4^-(aq)} + 5\mathrm{C_2O_4^{2-}(aq)} + 16\mathrm{H^+(aq)} \rightarrow 2\mathrm{Mn^{2+}(aq)} + 10\mathrm{CO_2(g)} + 8\mathrm{H_2O(l)}$ **Topic:** Volumetric Analysis (Redox Titration). **Detailed Concept:**

- **Standard Solution:** A solution whose concentration is accurately known. It is typically prepared by dissolving a precisely weighed amount of a primary standard substance in a known volume of solvent.
- **Stoichiometry in Titration:** The balanced chemical equation provides the mole ratio between the reactants. This ratio is crucial for calculating the unknown concentration. In this case, 2 moles of $\mathrm{MnO_4^-}$ react with 5 moles of $\mathrm{C_2O_4^{2-}}$. **Answer:** (i) A standard solution is a solution whose concentration is accurately known. (ii) **Step 1: Calculate moles of sodium oxalate ($\mathrm{Na_2C_2O_4}$).** Volume of $\mathrm{Na_2C_2O_4} = 25\ \mathrm{cm^3} = 0.025\ \mathrm{dm^3}$. Concentration of $\mathrm{Na_2C_2O_4} = 2.388\ \mathrm{M}$. Moles of $\mathrm{Na_2C_2O_4} = \mathrm{Concentration} \times \mathrm{Volume} = 2.388\ \mathrm{mol/dm^3} \times 0.025\ \mathrm{dm^3} = 0.0597\ \mathrm{mol}$.

Step 2: Use stoichiometry to find moles of $\mathrm{KMnO_4}$. From the balanced equation, $2\ \mathrm{moles}\ \mathrm{MnO_4^-}$ react with $5\ \mathrm{moles}\ \mathrm{C_2O_4^{2-}}$. Moles of $\mathrm{MnO_4^-} = (2/5) \times \mathrm{Moles\ of\ C_2O_4^{2-}} = (2/5) \times 0.0597\ \mathrm{mol} = 0.02388\ \mathrm{mol}$.

Step 3: Calculate the concentration of $\mathrm{KMnO_4}$. Volume of $\mathrm{KMnO_4} = 18\ \mathrm{cm^3} = 0.018\ \mathrm{dm^3}$. Concentration of $\mathrm{KMnO_4} = \mathrm{Moles/Volume} = 0.02388\ \mathrm{mol}/0.018\ \mathrm{dm^3} = 1.327\ \mathrm{M}$. **Why others are wrong:**

- Forgetting the $2:5$ mole ratio from the balanced equation is the most common error, leading to an incorrect number of moles for $\mathrm{KMnO_4}$.
- Failing to convert volumes from $\mathrm{cm^3}$ to $\mathrm{dm^3}$ before calculation will result in an incorrect concentration value.

Question 2(b)

Rewrite: X-rays with wavelength 1.54Å are incident on a crystal with interplanar spacing 3.35Å. Calculate the angle of incidence for first-order ($n = 1$) using the Bragg equation. **Topic:** Solid State Chemistry / X-ray Diffraction. **Detailed Concept:**

- **Bragg's Law:** Describes the conditions for constructive interference of X-rays diffracted by a crystal lattice. The formula is $n\lambda = 2d\sin\theta$, where:
  - n is the order of diffraction (an integer, given as 1 for first-order).
  - λ is the wavelength of the X-rays.
  - d is the interplanar spacing (distance between crystal planes).
  - θ is the angle of incidence (Bragg angle). **Answer:** Given: $n = 1$, $\lambda = 1.54\text{Å}$, $d = 3.35\text{Å}$. Using Bragg's Law: $n\lambda = 2d\sin\theta$ $1 \times 1.54\text{Å} = 2 \times 3.35\text{Å} \times \sin\theta$ $1.54 = 6.70 \times \sin\theta$ $\sin\theta = 1.54/6.70 = 0.22985$ $\theta = \arcsin(0.22985) = 13.29^\circ$. **Why others are wrong:**
- A common mistake is to forget the factor of 2 in $2d\sin\theta$, leading to an incorrect value for $\sin\theta$.
- Incorrectly converting units or failing to ensure all units are consistent (e.g., using different units for λ and d) can also lead to errors.

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SECTION B: INORGANIC CHEMISTRY

Question 3(a) & (b)

Rewrite: (a) What is meant by "periodic trend"? (b) (i) What are isoelectronic ions? (ii) List the isoelectronic anions of period 2. (iii) State and explain how the ionic radii of the

isoelectronic anions vary across the period 2 elements. **Topic:** Periodicity. **Detailed Concept:**

- **Periodic Trend:** Refers to the gradual and predictable changes in the physical and chemical properties of elements as one moves across a period or down a group in the periodic table. These trends are a consequence of the changing electron configurations and nuclear charge.
- **Isoelectronic Species:** Atoms or ions that have the same number of electrons and thus the same electron configuration. While they have the same electron count, their nuclear charges differ, which affects their size.
- **Ionic Radii:** The size of an ion. For isoelectronic species, the size is primarily determined by the nuclear charge. A higher nuclear charge pulls the electron cloud more tightly, resulting in a smaller ionic radius. **Answer:** (a) A **periodic trend** is the gradual change or repetition of chemical and physical properties of elements across a period or down a group in the periodic table. (b) (i) **Isoelectronic ions** are ions of different elements that possess the same number of electrons and thus the same electron configuration. (ii) The isoelectronic anions of period 2 (all having 10 electrons, like Neon) are: N^{3-}, O^{2-}, F^{-}. (iii) **Trend:** The ionic radius decreases across the period from N^{3-} to F^{-}. **Explanation:** All three ions (N^{3-}, O^{2-}, F^{-}) are isoelectronic, meaning they all have 10 electrons. However, they have different nuclear charges (number of protons):
- N^{3-} has 7 protons.
- O^{2-} has 8 protons.
- F^{-} has 9 protons. As the nuclear charge increases from Nitrogen to Fluorine, the greater positive charge in the nucleus exerts a stronger electrostatic attraction on the same number of electrons. This stronger pull draws the electron cloud closer to the nucleus, resulting in a decrease in ionic radius. **Why others are wrong:**
- Saying “same properties” for isoelectronic ions is incorrect; they only have the same electron configuration. Their chemical properties will differ due to different nuclear charges.
- Incorrectly stating that ionic radius increases across the period for isoelectronic anions, or failing to explain the role of increasing nuclear charge on the electron cloud, would be a common error.

Question 3©

Rewrite: Which element in period 2 has the highest electronegativity? **Topic:** Periodicity / Electronegativity. **Detailed Concept: Electronegativity** is the measure of the tendency of an atom to attract a bonding pair of electrons. It generally increases across a period due to increasing nuclear charge and decreasing atomic radius, leading to a stronger pull on valence electrons. **Answer:** The element in Period 2 with the highest electronegativity is **Fluorine (F)**. **Why others are wrong:**

- Some might incorrectly choose Oxygen or Nitrogen. While these are highly electronegative, Fluorine is the most electronegative element in the entire periodic table.
- Confusing electronegativity with electron affinity (energy change when an electron is added) or ionization energy (energy to remove an electron) is also a common mistake.

Question 3(d)

Rewrite: (i) What is a diatomic molecule? (ii) State the general outer electronic configuration exhibited by halogens. (iii) Write a balanced equation for the reaction of bromine with cold dilute NaOH. (iv) Type of reaction and definition. (v) Arrange the oxoacids of chlorine (HClO, $\mathrm{HClO_2}$, $\mathrm{HClO_3}$, $\mathrm{HClO_4}$) in terms of decreasing thermal stability. **Topic:** Group VII (Halogens) / Chemical Reactions / Acid Strength. **Detailed Concept:**

- **Diatomic Molecule:** A molecule composed of only two atoms, which can be of the same or different chemical elements.
- **Halogen Electronic Configuration:** Halogens are Group 17 elements, characterized by having seven valence electrons, one short of a stable octet.
- **Halogen-Alkali Reaction:** Halogens react with cold, dilute alkali solutions in a disproportionation reaction, where the halogen is simultaneously oxidized and reduced.
- **Thermal Stability of Oxoacids:** For oxoacids of the same central atom, thermal stability generally increases with increasing oxidation state of the central atom. This is because higher oxidation states lead to stronger bonds and more resonance stabilization. **Answer:** (i) A **diatomic molecule** is a molecule composed of only two atoms. (ii) The general outer electronic configuration

exhibited by halogens is ns^2np^5. (iii) Balanced equation for the reaction of bromine with cold dilute NaOH: $Br_2(aq) + 2NaOH(aq) \rightarrow NaBr(aq) + NaOBr(aq) + H_2O(l)$ (iv) This is a **disproportionation reaction**. **Definition:** A disproportionation reaction is a type of redox reaction in which a single element is simultaneously oxidized to a higher oxidation state and reduced to a lower oxidation state. (v) Arrangement of oxoacids of chlorine in terms of decreasing thermal stability: $HClO_4 > HClO_3 > HClO_2 > HClO$ **Why others are wrong:**

- Forgetting the state symbols in the chemical equation or balancing it incorrectly are common errors.
- Misidentifying the reaction type (e.g., calling it a simple redox reaction without specifying disproportionation) would be inaccurate.
- Incorrectly ordering the thermal stability of oxoacids, perhaps confusing it with acid strength (which follows the same trend but is a different property), is a frequent mistake.

Question 3(e)

Rewrite: (i) Draw a dot and cross diagram to illustrate the bonding in the formation of a nitrogen molecule from two nitrogen atoms. (ii) Using chemical equations only, outline the different steps involved in the industrial preparation of nitric acid. (iii) State one large-scale use of nitric acid. **Topic:** Chemical Bonding / Industrial Chemistry. **Detailed Concept:**

- **Nitrogen Molecule (N2):** Nitrogen atoms achieve a stable octet by sharing three pairs of electrons, forming a triple covalent bond. A dot and cross diagram visually represents the valence electrons and their sharing.
- **Ostwald Process (Industrial Nitric Acid):** This is a three-step industrial process for producing nitric acid from ammonia. It involves catalytic oxidation of ammonia, oxidation of nitric oxide, and absorption of nitrogen dioxide in water. **Answer:** (i) **Dot and Cross Diagram for N_2:**

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:N:::N:
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(Each N atom has one lone pair and shares three pairs of electrons with the other N atom).

(ii) **Industrial Preparation of Nitric Acid (Ostwald Process): Step 1: Catalytic oxidation of ammonia.** $4\mathrm{NH_3(g)} + 5\mathrm{O_2(g)} \xrightarrow{\mathrm{Pt/Rh\ catalyst,\ 800\text{-}900^\circ C}} 4\mathrm{NO(g)} + 6\mathrm{H_2O(g)}$

Step 2: Oxidation of nitric oxide to nitrogen dioxide. $2\mathrm{NO(g)} + \mathrm{O_2(g)} \rightarrow 2\mathrm{NO_2(g)}$

Step 3: Absorption of nitrogen dioxide in water. $3\mathrm{NO_2(g)} + \mathrm{H_2O(l)} \rightarrow 2\mathrm{HNO_3(aq)} + \mathrm{NO(g)}$ (The NO produced is recycled back to Step 2).

(iii) One large-scale use of nitric acid is in the **manufacture of fertilizers** (e.g., ammonium nitrate). **Why others are wrong:**

- Drawing an incorrect number of shared or lone pairs in the dot and cross diagram for $\mathrm{N_2}$ (e.g., showing a double or single bond) is a fundamental error.
- Missing any of the three key steps or providing incorrect reagents/conditions for the Ostwald process would result in loss of marks.
- Stating a minor or laboratory use instead of a large-scale industrial use for nitric acid would be incorrect.

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Question 4(a)

Rewrite: (i) Write the electronic configuration of potassium (K) and calcium ion ($\mathrm{Ca^{2+}}$). (ii) In the same period, the boiling points of Group II metals are generally higher than those of Group I metals. Explain. (iii) State and explain the trend in the solubility of Group I and Group II metal sulphates down each group. (iv) Why are the Group II metal hydroxides less soluble in water than the Group I metal hydroxides? **Topic:** Group I and II Metals / Periodicity / Solubility. **Detailed Concept:**

- **Electronic Configuration:** Describes the arrangement of electrons in an atom or ion. For ions, electrons are removed from or added to the highest energy levels.
- **Boiling Points of Metals:** Determined by the strength of metallic bonding, which depends on the number of delocalized electrons and the size of the metal ions.
- **Solubility of Sulfates:** For Group I, solubility generally increases down the group. For Group II, solubility generally decreases down the group. This is often explained by the interplay of lattice energy and hydration energy.

- **Solubility of Hydroxides:** For Group I, hydroxides are highly soluble. For Group II, solubility increases down the group. This is also related to lattice and hydration energies. **Answer:** (i) **Electronic Configuration:**
- Potassium (K): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ or $[\mathrm{Ar}] 4s^1$
- Calcium ion (Ca^{2+}): $1s^2 2s^2 2p^6 3s^2 3p^6$ or $[\mathrm{Ar}]$

(ii) **Explanation for higher boiling points of Group II metals:** Group II metals have generally higher boiling points than Group I metals in the same period because they have stronger metallic bonding. Group II metals contribute two valence electrons per atom to the delocalized electron sea, compared to Group I metals which contribute only one. This results in a stronger electrostatic attraction between the positively charged metal ions and the larger electron sea in Group II metals, requiring more energy to overcome during boiling.

(iii) **Trend in solubility of Group I and Group II metal sulphates:**

- **Group I Sulfates:** Solubility generally **increases** down the group (e.g., $\mathrm{Li_2SO_4}$ is less soluble than $\mathrm{Cs_2SO_4}$).
- **Group II Sulfates:** Solubility generally **decreases** down the group (e.g., $\mathrm{BeSO_4}$ is very soluble, while $\mathrm{BaSO_4}$ is insoluble). **Explanation:** The trend in solubility is determined by the relative magnitudes of lattice energy and hydration energy. For Group I sulfates, the decrease in lattice energy down the group (due to increasing ionic size) outweighs the decrease in hydration energy, leading to increased solubility. For Group II sulfates, the decrease in hydration energy (due to increasing ionic size of the cation) is more significant than the decrease in lattice energy, leading to decreased solubility.

(iv) **Why Group II metal hydroxides are less soluble than Group I metal hydroxides:** Group II metal hydroxides are generally less soluble than Group I metal hydroxides because the lattice energies of Group II hydroxides are significantly higher. Group II metal ions have a +2 charge and are smaller than Group I ions in the same period, leading to a stronger electrostatic attraction between the metal cation and the hydroxide anion. This results in a higher lattice energy that is harder to overcome by hydration energy, making them less soluble. **Why others are wrong:**

- Incorrectly writing electronic configurations, especially for ions, by not removing electrons from the highest energy level first.

- Failing to link boiling points to the strength of metallic bonding and the number of delocalized electrons.
- Confusing the solubility trends for Group I and Group II sulfates or hydroxides, or providing an incomplete explanation that doesn't consider both lattice and hydration energies.

Question 4(b)

Rewrite: (i) Explain what the term "inert pair effect" means. (ii) In the axis below, sketch the trend in the stability of the +2 oxidation state down the Group IV elements. (iii) Explain the trend indicated above. (iv) Write equations for the hydrolysis of $\mathrm{CCl_4}$ and $\mathrm{SiCl_4}$. **Topic:** Group IV Elements / Inert Pair Effect / Hydrolysis. **Detailed Concept:**

- **Inert Pair Effect:** The reluctance of the outermost s-electrons to participate in chemical bonding, leading to a stable oxidation state that is two less than the group valency. This effect becomes more pronounced down a group for heavier p-block elements.
- **Stability of Oxidation States in Group IV:** For Group IV elements, the +4 oxidation state is more stable for lighter elements (C, Si), while the +2 oxidation state becomes more stable for heavier elements (Ge, Sn, Pb) due to the inert pair effect.
- **Hydrolysis of Halides:** The reaction of a compound with water, often involving the breaking of bonds and formation of new ones with H and OH from water. The ability to hydrolyze depends on the availability of empty d-orbitals and the polarity of the bonds. **Answer:** (i) The **inert pair effect** is the tendency of the two outermost s-electrons in heavier p-block elements (especially in Groups 13-16) to remain unshared and un-ionized, leading to a stable oxidation state that is two less than the group valency. This effect increases down the group.

(ii) **Sketch of trend in stability of +2 oxidation state down Group IV:**



(The graph should show an increasing trend in the stability of the +2 oxidation state from C to Pb).

(iii) **Explanation of the trend:** As we move down Group IV from Carbon to Lead, the stability of the +2 oxidation state increases, while the stability of the +4 oxidation state decreases. This is due to the **inert pair effect**. For heavier elements like Tin (Sn) and Lead (Pb), the ns^2 electrons (where n is the principal quantum number) become increasingly reluctant to participate in bonding. This is attributed to poor shielding by inner d and f electrons, which leads to a stronger attraction of the ns^2 electrons to the nucleus, making them less available for bond formation. Consequently, the elements prefer to form compounds by using only their two np^2 electrons, resulting in a more stable +2 oxidation state.

(iv) **Equations for the hydrolysis of $\mathrm{CCl_4}$ and $\mathrm{SiCl_4}$:**

- **Hydrolysis of $\mathrm{CCl_4}$:** $\mathrm{CCl_4}$ does **not** hydrolyze under normal conditions.
  - **Explanation:** Carbon does not have empty d-orbitals to accept the lone pair of electrons from the oxygen atom of water, which is necessary to form an intermediate before hydrolysis can occur. Also, the C-Cl bond is strong.

- **Hydrolysis of $\mathrm{SiCl_4}$:** $\mathrm{SiCl_4}$ undergoes rapid hydrolysis. $\mathrm{SiCl_4(l) + 2H_2O(l) \rightarrow SiO_2(s) + 4HCl(aq)}$
  - **Explanation:** Silicon has vacant 3d-orbitals that can accept the lone pair of electrons from the oxygen atom of water, forming an intermediate. This facilitates the nucleophilic attack by water and subsequent hydrolysis. **Why others are wrong:**

- Incorrectly defining the inert pair effect or misapplying it to elements outside the p-block.
- Drawing a decreasing trend for the +2 oxidation state stability or failing to explain the underlying reasons (poor shielding, nuclear attraction).
- Incorrectly stating that CCl_4 hydrolyzes or failing to explain the difference in hydrolysis behavior between CCl_4 and $SiCl_4$ based on the availability of d-orbitals.

Question 4©

Rewrite: Consider the first row d-block elements (Scandium to Zinc). Which of these elements is/are not transition metals? Explain. **Topic:** Transition Metals. **Detailed Concept:** A **transition metal** is defined as an element that has an incompletely filled d-subshell in its atomic state or in any of its common ions. This definition excludes elements like Scandium (sometimes) and Zinc. **Answer:** The element that is **not** considered a transition metal among the first-row d-block elements (Scandium to Zinc) is **Zinc (Zn)**. **Explanation:**

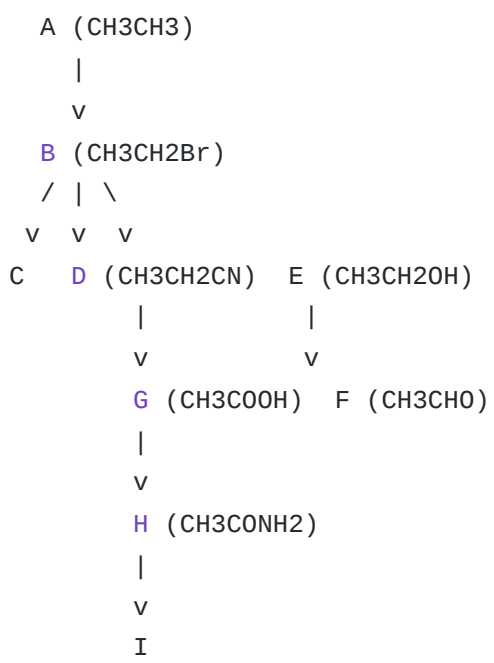
- **Zinc (Zn):** Its electronic configuration is $[Ar]3d^{10}4s^2$. In its most common and stable oxidation state, Zn^{2+}, its electronic configuration is $[Ar]3d^{10}$. In both the atomic state and its common ion, Zinc has a completely filled d-subshell. Therefore, it does not meet the definition of a transition metal.
- **Scandium (Sc):** While Scandium (Sc) has an electronic configuration of $[Ar]3d^14s^2$, its only common ion is Sc^{3+}, which has an electronic configuration of $[Ar]$ (no d-electrons). Some definitions might exclude Scandium based on its ion not having an incompletely filled d-subshell. However, the IUPAC definition typically focuses on the atomic state or *any* common ion. Given the options, Zinc is the clearer non-transition metal. **Why others are wrong:**
- Incorrectly including Zinc as a transition metal, or excluding elements like Copper (which has a d^{10} configuration in its atomic state but a d^9 in Cu^{2+} ion, making it a transition metal).
- Failing to provide the electronic configurations of the element and its common ion to justify the answer based on the definition of a transition metal.

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SECTION C: ORGANIC CHEMISTRY

Question 5(a)

Rewrite: Study the following reaction scheme and answer the questions that follow:



(i) Write the names and structural formulae of compounds C and I. (ii) Give the reagents and reaction conditions necessary for the following conversions:

Steps	Reagents	Reaction Conditions
(i)		
(ii)		
(iii)		
(iv)		
(v)		

(iii) Compare the procedures for converting E to F and E to G.

Topic: Organic Synthesis / Reaction Mechanisms. **Detailed Concept:** This question tests knowledge of various organic reactions, including free radical substitution, nucleophilic substitution, Grignard reagent formation, nitrile synthesis, oxidation of alcohols, and reduction of amides.

Answer: (i) **Names and Structural Formulae of C and I:**

- **Compound C:** Formed from B (Bromoethane, CH_3CH_2Br). This is likely the formation of a Grignard reagent.
    - **Name:** Ethylmagnesium bromide
    - **Structural Formula:** CH_3CH_2MgBr
- **Compound I:** Formed from H (Ethanamide, CH_3CONH_2). This is likely the reduction of an amide to an amine.
    - **Name:** Ethanamine (or Ethylamine)
    - **Structural Formula:** $CH_3CH_2NH_2$

(ii) **Reagents and Reaction Conditions for Conversions:**

Steps	Reagents	Reaction Conditions
(i)	Br_2	UV light / Heat (Free radical substitution)
(ii)	Mg	Dry ether (Formation of Grignard reagent)
(iii)	KCN (or $NaCN$)	Ethanolic solution, Heat (Nucleophilic substitution)
(iv)	$LiAlH_4$ (or H_2/Ni)	Dry ether, followed by H_3O^+ (Reduction of nitrile to amine)
(v)	$LiAlH_4$ (or H_2/Ni)	Dry ether, followed by H_3O^+ (Reduction of amide to amine)

(iii) **Comparison of procedures for converting E to F and E to G:**

- **Conversion of E (Ethanol, CH_3CH_2OH) to F (Ethanal, CH_3CHO):** This is the **partial oxidation** of a primary alcohol to an aldehyde. It requires a mild oxidizing agent and careful control of conditions to prevent further oxidation to the carboxylic acid. Typical reagents include acidified potassium dichromate(VI) ($K_2Cr_2O_7/H^+$) or pyridinium chlorochromate (PCC), with immediate distillation of the aldehyde as it forms.

- **Conversion of E (Ethanol, CH_3CH_2OH) to G (Ethanoic acid, CH_3COOH):** This is the **complete oxidation** of a primary alcohol to a carboxylic acid. It requires a strong oxidizing agent and heating under reflux to ensure complete oxidation. Typical reagents include hot acidified potassium dichromate(VI) ($K_2Cr_2O_7/H^+$) or hot acidified potassium permanganate(VII) ($KMnO_4/H^+$).

Key Differences:

- **Extent of Oxidation:** Partial vs. Complete.
- **Oxidizing Agent Strength:** Mild vs. Strong.
- **Reaction Conditions:** Immediate distillation vs. Heating under reflux.

Why others are wrong:

- **For (i):** Incorrectly identifying C as a simple alkane or I as a nitrile. The Grignard reagent formation and amide reduction are specific reactions.
- **For (ii):** Providing incorrect reagents or conditions (e.g., using aqueous $NaOH$ for Grignard formation, which would react with water). Forgetting UV light for free radical substitution or using the wrong solvent for KCN reaction.
- **For (iii):** Failing to distinguish between partial and complete oxidation of primary alcohols, or using the same reagents/conditions for both conversions. Not mentioning the need for immediate distillation for aldehyde formation or reflux for carboxylic acid formation.

Question 5(b)

Rewrite: Suggest a chemical test to distinguish between: (i) Phenol and Benzyl alcohol (ii) Ethylamine and Acetamide (iii) Acetone and Methyl acetate

Topic: Functional Group Analysis / Qualitative Tests. **Detailed Concept:** This question requires knowledge of characteristic reactions of different functional groups to devise simple chemical tests that yield distinct observable results.

Answer: (i) **Phenol and Benzyl alcohol:**

- **Test:** Add neutral Iron(III) chloride ($FeCl_3$) solution.
- **Observation:**
    - **Phenol:** Gives a characteristic violet, blue, or green coloration (due to the formation of a complex with the phenolic -OH group).

    - **Benzyl alcohol:** No visible color change.
- **Why others are wrong:** Using sodium metal would not distinguish them as both have an -OH group and would react to produce hydrogen gas. Using a test for primary alcohols (like oxidation with acidified dichromate) would also not be ideal as phenol is not an alcohol.

(ii) **Ethylamine ($CH_3CH_2NH_2$) and Acetamide (CH_3CONH_2):**

- **Test:** Add dilute hydrochloric acid (HCl) followed by warming, then add $NaOH$ and warm.
- **Observation:**
    - **Ethylamine:** Being a base, it will react with HCl to form a salt. No ammonia gas is evolved upon warming with $NaOH$.
    - **Acetamide:** Upon warming with dilute HCl (hydrolysis), it will produce ammonium chloride and ethanoic acid. Upon warming with $NaOH$ (hydrolysis), it will produce sodium acetate and **ammonia gas** (which turns red litmus blue).
- **Alternative Test:** Reaction with nitrous acid (HNO_2, generated in situ from $NaNO_2$ and dilute HCl).
    - **Ethylamine (primary amine):** Reacts with nitrous acid to produce nitrogen gas (effervescence) and an alcohol.
    - **Acetamide (amide):** Does not react with nitrous acid in the same way to produce nitrogen gas. It might hydrolyze slowly.
- **Why others are wrong:** Simply testing for basicity with litmus paper might not be conclusive enough as both can show some basic character (though ethylamine is a stronger base). The hydrolysis of amide to produce ammonia is a key distinguishing feature.

(iii) **Acetone (Propanone, CH_3COCH_3) and Methyl acetate (Methyl ethanoate, CH_3COOCH_3):**

- **Test:** Iodoform test (reaction with iodine in alkaline solution).
- **Observation:**
    - **Acetone:** Gives a positive iodoform test, producing a yellow precipitate of triiodomethane (iodoform, CHI_3) and a characteristic antiseptic smell. This is because acetone contains the CH_3CO- group.

- **Methyl acetate:** Gives a negative iodoform test (does not contain the CH_3CO- group directly attached to a hydrogen or carbon that can be oxidized to a carbonyl).

- **Why others are wrong:** Using Tollens' reagent or Fehling's solution would not work as neither is an aldehyde. Simple tests for carbonyls (like 2,4-DNPH) would be positive for acetone but not for methyl acetate (an ester), but the iodoform test is more specific for methyl ketones.

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Question 6(a)

Rewrite: Give the formulas of two organic compounds that would react together to give each of the following products. (i) CH_3COOCH_3 (Methyl acetate) (ii) $CH_3CH_2NHCOCH_3$ (N-ethylacetamide)

Topic: Organic Reactions / Synthesis. **Detailed Concept:** This question requires identifying the reactants for esterification and amide formation. Esters are typically formed from a carboxylic acid and an alcohol. Amides can be formed from an acyl chloride (or carboxylic acid) and an amine.

Answer: (i) **Product:** CH_3COOCH_3 **(Methyl acetate)** This is an ester. It can be formed from a carboxylic acid and an alcohol.

- **Reactants:**
  1. Ethanoic acid (Acetic acid): CH_3COOH
  2. Methanol: CH_3OH (Reaction: $CH_3COOH + CH_3OH \xrightarrow{\text{Conc. } H_2SO_4, \text{heat}} CH_3COOCH_3 + H_2O$)

(ii) **Product:** $CH_3CH_2NHCOCH_3$ **(N-ethylacetamide)** This is an N-substituted amide. It can be formed from an acyl chloride (or carboxylic acid/anhydride) and a primary amine.

- **Reactants:**
  1. Ethanoyl chloride (Acetyl chloride): CH_3COCl
  2. Ethylamine: $CH_3CH_2NH_2$ (Reaction: $CH_3COCl + CH_3CH_2NH_2 \rightarrow CH_3CH_2NHCOCH_3 + HCl$)

- **Alternative Reactants:** Ethanoic acid (CH_3COOH) and Ethylamine ($CH_3CH_2NH_2$) can also form this amide, usually by heating the salt formed initially.

Why others are wrong:

- For (i), incorrectly identifying the alcohol or carboxylic acid (e.g., using ethanol instead of methanol for methyl acetate).
- For (ii), using an incorrect amine or acylating agent. Forgetting that a primary amine is needed for an N-substituted amide, or using a secondary amine which would lead to a different product.

Question 6(b)

Rewrite: Provide the systematic names for the following amines. (i) CH_3NHCH_3 (ii) $C_6H_5CH_2NH_2$ (iii) $C_6H_5NHCH_3$ (iv) $C_6H_5NH_2$

Topic: IUPAC Nomenclature of Amines. **Detailed Concept:**

- **Primary Amines:** Named by replacing the '-e' of the corresponding alkane with '-amine' or by naming the alkyl group followed by 'amine'.
- **Secondary/Tertiary Amines:** Named as N-substituted primary amines, where the largest alkyl group is considered the parent amine, and other alkyl groups are designated as 'N-alkyl'.
- **Aromatic Amines:** Often have common names like aniline, but systematic names are also used. **Answer:** (i) CH_3NHCH_3: **N-methylmethanamine** (or Dimethylamine) (ii) $C_6H_5CH_2NH_2$: **Phenylmethanamine** (or Benzylamine) (iii) $C_6H_5NHCH_3$: **N-methylphenylamine** (or N-methylaniline) (iv) $C_6H_5NH_2$: **Phenylamine** (or Aniline) **Why others are wrong:**
- Confusing primary, secondary, and tertiary amine nomenclature. Forgetting the 'N-' prefix for substituted amines.
- Using common names when systematic names are expected, or vice-versa if not specified. Incorrectly naming the phenyl group as a substituent on an alkyl chain or vice-versa.

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Question 6©

Rewrite: State and explain which of the two amines (phenylmethyl)amine and phenylamine is a weaker base. **Topic:** Basicity of Amines. **Detailed Concept:** The basicity of an amine depends on the availability of the lone pair of electrons on the nitrogen atom for donation. Factors affecting this include inductive effects (electron-donating groups increase basicity) and resonance effects (delocalization of the lone pair into an aromatic ring decreases basicity).

Answer: Phenylamine (Aniline, $C_6H_5NH_2$) is a weaker base than phenylmethylamine (Benzylamine, $C_6H_5CH_2NH_2$).

Explanation:

- **Phenylamine ($C_6H_5NH_2$):** In phenylamine, the lone pair of electrons on the nitrogen atom is delocalized into the benzene ring through resonance. This delocalization makes the lone pair less available for protonation, thus reducing its basicity. The resonance structures show the lone pair being shared with the ring, making the nitrogen less electron-rich.
- **Phenylmethylamine ($C_6H_5CH_2NH_2$):** In phenylmethylamine, the lone pair of electrons on the nitrogen atom is not directly conjugated with the benzene ring because of the intervening $-CH_2-$ group. The lone pair is localized on the nitrogen and is readily available for protonation. The phenyl group, being further away, has a much weaker inductive effect on the nitrogen's basicity compared to the direct resonance effect in phenylamine.

Therefore, because the lone pair on nitrogen in phenylamine is less available due to resonance delocalization, phenylamine is a weaker base. **Why others are wrong:**

- Incorrectly stating that phenylmethylamine is weaker, or failing to explain the role of resonance in reducing the basicity of aromatic amines.
- Confusing inductive effects with resonance effects, or not clearly explaining how electron delocalization affects the availability of the lone pair.

Question 6(d)

Rewrite: Give the reagents and reaction conditions for the conversion of: (i) $CH_3CH(Br)CH_3$ to $CH_3CH(OH)CH_3$ (ii) CH_3CH_2Br to CH_3CH_3 (iii) Name the type of reaction mechanism taking place and state the role of the reagent in the formation of

the product in d(ii) above. (iv) Outline the mechanism for the conversion of $CH_3CH(Br)CH_3$ to $CH_3CH(OH)CH_3$.

Topic: Organic Reactions / Mechanisms (Nucleophilic Substitution, Reduction). **Detailed Concept:** This question covers nucleophilic substitution (SN1/SN2) for alcohol formation and reduction for alkane formation, along with mechanism details.

Answer: (i) **Conversion of $CH_3CH(Br)CH_3$ (2-Bromopropane) to $CH_3CH(OH)CH_3$ (Propan-2-ol):**

- **Reagents:** Aqueous potassium hydroxide (KOH) or sodium hydroxide (NaOH).
- **Reaction Conditions:** Warm (heat under reflux). (This is a nucleophilic substitution reaction where OH^- replaces Br^-.)

(ii) **Conversion of CH_3CH_2Br (Bromoethane) to CH_3CH_3 (Ethane):**

- **Reagents:** Lithium aluminium hydride ($LiAlH_4$) or Hydrogen gas (H_2) with a catalyst (e.g., Ni, Pt, Pd).
- **Reaction Conditions:** For $LiAlH_4$: Dry ether, followed by aqueous acid workup. For H_2: Heat, pressure, catalyst. (This is a reduction reaction.)

(iii) **Mechanism and Role of Reagent for d(ii):**

- **Type of Reaction Mechanism:** Reduction (specifically, nucleophilic substitution followed by hydride transfer if using $LiAlH_4$, or catalytic hydrogenation).
- **Role of Reagent (e.g., $LiAlH_4$):** $LiAlH_4$ acts as a **reducing agent**, providing hydride ions (H^-) which are strong nucleophiles that replace the bromine atom, effectively reducing the haloalkane to an alkane.

(iv) **Mechanism for the conversion of $CH_3CH(Br)CH_3$ to $CH_3CH(OH)CH_3$:** This is the hydrolysis of a secondary haloalkane, which can proceed via both S_N1 and S_N2 mechanisms, but S_N1 is often favored for secondary haloalkanes in protic solvents (like water).

S_N1 **Mechanism (Two steps): Step 1: Ionization (slow, rate-determining step)** The carbon-bromine bond breaks heterolytically, forming a stable secondary carbocation and a bromide ion. $CH_3CH(Br)CH_3 \rightarrow CH_3CH^+CH_3 + Br^-$

Step 2: Nucleophilic attack (fast step) The nucleophile (water, H_2O, or hydroxide ion, OH^- if using $NaOH$) attacks the carbocation. If water attacks, a protonated alcohol is

formed, which then loses a proton to yield the final alcohol. $\mathrm{CH_3CH^+CH_3} + \mathrm{H_2O} \rightarrow \mathrm{CH_3CH(OH_2^+)CH_3} \rightarrow \mathrm{CH_3CH(OH)CH_3} + \mathrm{H^+}$

Why others are wrong:

- For (i) and (ii), using incorrect reagents (e.g., alcoholic KOH for (i) would lead to elimination, not substitution) or conditions.
- For (iii), misidentifying the mechanism or the role of the reagent (e.g., calling $\mathrm{LiAlH_4}$ an oxidizing agent).
- For (iv), incorrectly drawing the mechanism (e.g., showing a one-step $\mathrm{S_N2}$ for a secondary haloalkane in aqueous conditions, or missing the carbocation intermediate in $\mathrm{S_N1}$). Not showing the correct bond breaking/forming or electron movement.

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Question 6(e)

Rewrite: Acylation of the benzene ring can occur with an acyl chloride by the reaction: $\mathrm{C_6H_6} + \mathrm{CH_3COCl} \xrightarrow{\mathrm{AlCl_3}} \mathrm{Y} + \mathrm{HCl}$ (i) Draw the structure of the product Y. (ii) State the type of organic reaction mechanism taking place in the above reaction and draw the structure of the electrophilic reagent in the acylation of benzene.

Topic: Electrophilic Aromatic Substitution / Friedel-Crafts Acylation. **Detailed Concept:** This is a Friedel-Crafts acylation reaction, a type of electrophilic aromatic substitution. The acyl chloride reacts with a Lewis acid catalyst (like $\mathrm{AlCl_3}$) to generate a highly reactive acylium ion, which acts as the electrophile.

Answer: (i) **Structure of product Y:** Product Y is acetophenone (phenylethanone).



(Benzene ring with a $-COCH_3$ group attached).

(ii) **Type of reaction mechanism and structure of electrophilic reagent:**

- **Type of organic reaction mechanism: Electrophilic Aromatic Substitution** (specifically, Friedel-Crafts Acylation).
- **Structure of the electrophilic reagent:** The electrophile is the **acylium ion**.

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CH3
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(A carbon atom with a positive charge, double-bonded to oxygen, and single-bonded to a methyl group).

Why others are wrong:

- For (i), drawing an incorrect product (e.g., an alkyl benzene if confusing with Friedel-Crafts alkylation, or an alcohol/ester).
- For (ii), misidentifying the mechanism (e.g., nucleophilic substitution) or incorrectly drawing the electrophile (e.g., drawing CH_3COCl itself as the electrophile, or a carbocation instead of an acylium ion).

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